

# Notes: Altonji & Pierret (QJE, 2001)

## Employer Learning and Statistical Discrimination

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# 1 Big picture

- **Question.** Do employers statistically discriminate among young workers using easily observed traits, and does that reliance decline as employers learn workers' true productivity?
- **Core prediction.** If employers use an observable variable  $s$  such as schooling because it predicts an initially unobserved productivity variable  $z$  such as AFQT, then the wage coefficient on  $s$  should fall with experience while the wage coefficient on  $z$  should rise.
- **Empirical implementation.** The authors use NLSY79 data on young men and estimate wage equations with interactions between experience and schooling, AFQT, sibling wages, father's education, and race.
- **Main result for education.** The coefficient on AFQT, sibling wages, and other hard-to-observe productivity correlates rises with experience, while the schooling coefficient declines. This supports employer learning with statistical discrimination on education.
- **Main result for race.** The evidence gives little support for the idea that employers make full use of race as statistical information in wage setting. The race wage gap is small at labor-market entry and widens with experience, consistent with firms learning productivity-related information that is legally usable, but the authors emphasize important caveats.
- **Training.** Adding training controls does not eliminate the learning pattern. Training appears partly related to productivity beliefs, so the paper cannot cleanly separate employer learning from human capital accumulation.
- **Why it matters.** The paper gives a testable empirical strategy for a classic but hard-to-identify issue: whether a correlation between wages and an observable characteristic reflects direct productivity, statistical discrimination, or information updating.

# 2 Incremental contribution

- **From employer learning to statistical discrimination.** Farber and Gibbons show that the coefficient on initially unobserved productivity information should rise with experience. Altonji and Pierret add the key comparative prediction that the coefficient on the initially observable variable used for statistical discrimination should fall once the hard-to-observe correlate is allowed to gain weight.
- **A solution to a core identification problem.** A static wage regression cannot tell whether schooling receives a wage premium because it raises productivity, signals productivity, or proxies for omitted information. The paper's contribution is to use *changes in coefficients with experience* to distinguish these stories.
- **A general test, not only a labor-market application.** The logic applies whenever agents price or evaluate others under incomplete information and learn over time: labor markets, credit markets, insurance markets, and relationships between firms and customers.
- **Race versus education.** The paper applies the same logic to race and education but treats them differently. Education is legally and commonly used by employers; race is legally prohibited as information. This distinction makes the race evidence more subtle and provides a useful template for studying discriminatory inference.
- **Bridge between theory and wage dynamics.** The paper shows that experience interactions in wage regressions can be interpreted not only as human capital or training effects but also as the time path of employer information.
- **Research agenda contribution.** It shifts attention from whether employers have information to when they get it, how it changes compensation, and which variables are initially used to form beliefs.

## 3 Data and measurement

### 3.1 NLSY79 sample

- The empirical analysis uses the 1992 release of the National Longitudinal Survey of Youth 1979.
- The sample is restricted to men who are white or black and who have at least eight years of schooling.
- Labor-market observations before leaving school are excluded.
- The main wage outcome is log real hourly wage.
- The main sample contains 21,058 wage observations from 2,976 individuals.

**Interpretation:** The panel nature of the NLSY79 is essential because the test uses wage-experience profiles, not only cross-sectional wage differences.

### 3.2 Observable and hard-to-observe variables

- $s$ : an easily observable characteristic available to employers and the econometrician. The paper's main  $s$  variable is schooling.
- $z$ : a productivity correlate not directly observed by employers but observed by the econometrician. Main  $z$  variables are AFQT, the wage of an older sibling, and father's education.
- Race is observable, but the interpretation differs because employers are legally constrained from using race as information.
- The authors control for calendar-year effects, occupation of first job, urban residence, missing-information dummies, and interactions between key variables and calendar time.

**Interpretation:** The empirical design asks whether wage determination gradually shifts from public credentials toward deeper productivity correlates as labor-market information accumulates.

### 3.3 Experience measures

- Potential experience is age minus schooling minus six.
- Actual experience is the number of weeks worked more than thirty hours, divided by fifty.
- The main OLS results use potential experience.
- Results using actual experience instrumented by potential experience are also reported.

**Interpretation:** Potential experience is cleaner but less precise; actual experience is closer to work history but endogenous. The paper shows that the main pattern is robust to both.

## 4 Model and empirical specifications

### 4.1 Productivity decomposition

The log productivity of worker  $i$  with experience  $t_i$  is

$$y_{it} = rs_i + \alpha q_i + \lambda z_i + \eta_i + H(t_i). \quad (1)$$

- $s_i$ : observed by the employer and econometrician, e.g., schooling.
- $q_i$ : observed by the employer but not the econometrician.

- $z_i$ : observed by the econometrician but not initially by the employer, e.g., AFQT.
- $\eta_i$ : unobserved by both employer and econometrician.
- $H(t_i)$ : experience profile of productivity.

**Interpretation:** Employers form initial beliefs using  $s$  and  $q$  but do not initially observe  $z$  and  $\eta$ . The econometrician can use  $z$  to test whether employers learn.

## 4.2 Initial beliefs and learning

Employers form conditional expectations:

$$z = E(z|s, q) + v = \gamma_1 q + \gamma_2 s + v, \quad (2)$$

$$\eta = E(\eta|s, q) + e = \delta s + e. \quad (3)$$

The initial productivity-assessment error is  $\lambda v + e$ . Employers observe noisy signals of performance over time and update beliefs about this term.

**Interpretation:** If schooling predicts AFQT, then schooling receives part of the wage credit for AFQT at labor-market entry. As employers learn, AFQT receives more of that credit directly.

## 4.3 Wage equation implied by learning

The reduced-form conditional expectation can be written as

$$E(w_t|s, z, t) = b_{st}s + b_{zt}z + H^*(t). \quad (4)$$

Employer learning with statistical discrimination implies

$$\frac{\partial b_{zt}}{\partial t} \geq 0, \quad \frac{\partial b_{st}}{\partial t} \leq 0. \quad (5)$$

**Interpretation:** The coefficient on  $z$  rises because employers gradually discover information correlated with  $z$ . The coefficient on  $s$  falls because  $s$  was initially proxying for  $z$ .

## 4.4 Coefficient restriction

In the scalar case, the model implies

$$\frac{\partial b_{st}}{\partial t} = -\psi_{zs} \frac{\partial b_{zt}}{\partial t}, \quad (6)$$

where  $\psi_{zs}$  is the coefficient from regressing  $z$  on  $s$ .

**Interpretation:** The decline in the schooling coefficient should be mechanically linked to the rise in the coefficient on AFQT or other  $z$  variables. This is the paper's central overidentifying logic.

## 4.5 Empirical wage equation

The empirical specification has the form

$$w_{it} = \beta_0 + \beta_s s_i + \beta_z z_i + \beta_{sE}(s_i \times t_i/10) + \beta_{zE}(z_i \times t_i/10) + X'_{it}\Gamma + u_{it}. \quad (7)$$

- The hypothesis predicts  $\beta_{zE} > 0$  and  $\beta_{sE} < 0$ .
- The education coefficient should fall once  $z \times \text{experience}$  is included.
- Race specifications test whether Black behaves like an observable  $s$  variable or an initially unobserved  $z$  variable.

**Interpretation:** The identifying variation comes from how the effects of productivity correlates evolve with labor-market experience, after controlling for general experience profiles and secular trends.

## 5 Identification and interpretation

### 5.1 What makes the test informative

- If schooling is observable and AFQT is not initially observed by employers, schooling partly proxies for AFQT early in the career.
- Once employers learn, wage setting should respond more to actual productivity signals and less to schooling.
- Therefore, the experience path of coefficients can reveal statistical discrimination even when a static wage regression cannot.

**Interpretation:** The test uses the dynamics of information. The key is not whether schooling and AFQT predict wages, but whether their relative predictive power shifts over time.

### 5.2 Main threats to interpretation

- Schooling and AFQT may affect training and career paths, not only initial beliefs.
- Returns to schooling and AFQT may change across cohorts or calendar years.
- Employers may observe some information correlated with AFQT before hiring, which would weaken the clean interpretation.
- Race-related results are especially difficult because discrimination, premarket skill formation, employer information, networks, and training may all interact.

**Interpretation:** The paper is careful not to claim a fully structural decomposition between learning and human capital. Its strongest evidence is the sign pattern across multiple hard-to-observe productivity correlates.

## 6 Tables and exhibits

### 6.1 Table I: AFQT, schooling, and wages

**Read:**

- Panel 1 uses potential experience. Without AFQT-experience interactions, schooling has a coefficient of 0.0586 and AFQT has a coefficient of 0.0834.
- Adding  $\text{AFQT} \times \text{experience}$  makes the AFQT main effect near zero at entry, but the AFQT-experience coefficient is 0.0752.
- The schooling-experience coefficient falls from -0.0032 to -0.0234 when  $\text{AFQT} \times \text{experience}$  is added.
- Panel 2 uses actual experience instrumented by potential experience. The same qualitative pattern appears: AFQT gains importance with experience and the education slope becomes more negative.

**Economic message:** The main result appears immediately. Early wages rely more on schooling, while later wages rely more on AFQT. This is exactly the employer-learning-with-statistical-discrimination prediction.

**Interpretation:** The table should not be read as saying schooling is unproductive. Rather, conditional on the controls, the part of schooling that proxies for initially hidden productivity loses wage weight as employers observe workers directly. The race coefficients in this table also preview the paper’s race result: the Black wage gap is small at entry in some specifications but widens with experience.

TABLE I  
THE EFFECTS OF STANDARDIZED AFQT AND SCHOOLING ON WAGES  
Dependent Variable: Log Wage; OLS estimates (standard errors).

| Panel 1—Experience measure: potential experience                                      |                     |                     |                     |                     |
|---------------------------------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|
| Model:                                                                                | (1)                 | (2)                 | (3)                 | (4)                 |
| (a) Education                                                                         | 0.0586<br>(0.0118)  | 0.0829<br>(0.0150)  | 0.0638<br>(0.0120)  | 0.0785<br>(0.0153)  |
| (b) Black                                                                             | −0.1565<br>(0.0256) | −0.1553<br>(0.0256) | 0.0001<br>(0.0621)  | −0.0565<br>(0.0723) |
| (c) Standardized AFQT                                                                 | 0.0834<br>(0.0144)  | −0.0060<br>(0.0360) | 0.0831<br>(0.0144)  | 0.0221<br>(0.0421)  |
| (d) Education *<br>experience/10                                                      | −0.0032<br>(0.0094) | −0.0234<br>(0.0123) | −0.0068<br>(0.0095) | −0.0193<br>(0.0127) |
| (e) Standardized AFQT *<br>experience/10                                              |                     | 0.0752<br>(0.0286)  |                     | 0.0515<br>(0.0343)  |
| (f) Black * experience/10                                                             |                     |                     | −0.1315<br>(0.0482) | −0.0834<br>(0.0581) |
| $R^2$                                                                                 | 0.2861              | 0.2870              | 0.2870              | 0.2873              |
| Panel 2—Experience measure: actual experience instrumented<br>by potential experience |                     |                     |                     |                     |
| Model:                                                                                | (1)                 | (2)                 | (3)                 | (4)                 |
| (a) Education                                                                         | 0.0836<br>(0.0208)  | 0.1218<br>(0.0243)  | 0.0969<br>(0.0206)  | 0.1170<br>(0.0248)  |
| (b) Black                                                                             | −0.1310<br>(0.0261) | −0.1306<br>(0.0260) | 0.0972<br>(0.0851)  | 0.0178<br>(0.1029)  |
| (c) Standardized AFQT                                                                 | 0.0925<br>(0.0143)  | −0.0361<br>(0.0482) | 0.0881<br>(0.0143)  | 0.0062<br>(0.0572)  |
| (d) Education *<br>experience/10                                                      | −0.0539<br>(0.0235) | −0.0952<br>(0.0276) | −0.0665<br>(0.0234) | −0.0889<br>(0.0283) |
| (e) Standardized AFQT *<br>experience/10                                              |                     | 0.1407<br>(0.0514)  |                     | 0.0913<br>(0.0627)  |
| (f) Black * experience/10                                                             |                     |                     | −0.2670<br>(0.0968) | −0.1739<br>(0.1184) |
| $R^2$                                                                                 | 0.3056              | 0.3063              | 0.3061              | 0.3064              |

Experience is modeled with a cubic polynomial. All equations control for year effects, education interacted with a cubic time trend, Black interacted with a cubic time trend, AFQT interacted with a cubic time trend, two-digit occupation at first job, and urban residence. For these time trends, the base year is 1992. For the model in Panel 1 column (1) the coefficient on AFQT and Black are .0312 and −.1006, respectively, when evaluated for 1983. In Panel 2 the instrumental variables are the corresponding terms involving potential experience and the other variables in the model. Standard errors are White/Huber standard errors computed accounting for the fact that there are multiple observations for each worker. The sample size is 21,058 observations from 2976 individuals.

Table I: AFQT and schooling on wages.

TABLE II  
THE EFFECTS OF FATHER’S EDUCATION, SIBLING WAGES, AND SCHOOLING ON WAGES  
Dependent Variable: Log Wage; Experience Measure: Potential Experience.  
OLS estimates (standard errors)

| Model:                                       | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|----------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| (a) Education                                | 0.0611<br>(0.0160)  | 0.0620<br>(0.0166)  | 0.0568<br>(0.0163)  | 0.0658<br>(0.0167)  | 0.0666<br>(0.0129)  | 0.0730<br>(0.0140)  | 0.0704<br>(0.0130)  | 0.0734<br>(0.0140)  |
| (b) Black                                    | −0.2074<br>(0.0276) | −0.2076<br>(0.0276) | −0.0509<br>(0.0846) | −0.0878<br>(0.0871) | −0.2212<br>(0.0250) | −0.2209<br>(0.0250) | −0.0705<br>(0.0668) | −0.0793<br>(0.0692) |
| (c) Log of sibling’s wage                    | 0.1802<br>(0.0528)  | −0.0260<br>(0.0913) | 0.1817<br>(0.0329)  | 0.0010<br>(0.0940)  |                     |                     |                     |                     |
| (d) Father’s education/10                    |                     |                     |                     |                     | 0.0826<br>(0.0366)  | −0.0187<br>(0.1000) | 0.0829<br>(0.0364)  | 0.0314<br>(0.1030)  |
| (e) Education *<br>experience/10             | 0.0107<br>(0.0131)  | 0.0012<br>(0.0136)  | 0.0065<br>(0.0133)  | −0.0008<br>(0.0136) | 0.0023<br>(0.0104)  | −0.0029<br>(0.0113) | −0.0002<br>(0.0105) | −0.0027<br>(0.0113) |
| (f) Log of sibling’s wage *<br>experience/10 |                     | 0.1796<br>(0.0749)  |                     | 0.1571<br>(0.0770)  |                     |                     |                     |                     |
| (g) Father’s education *<br>experience/10    |                     |                     |                     |                     |                     | 0.0867<br>(0.0813)  |                     | 0.0441<br>(0.0841)  |
| (h) Black * experience/10                    |                     |                     | −0.1311<br>(0.0686) | −0.1004<br>(0.0704) |                     |                     | −0.1270<br>(0.0541) | −0.1194<br>(0.0563) |
| $R^2$                                        | 0.3183              | 0.3196              | 0.3191              | 0.3200              | 0.2748              | 0.2750              | 0.2755              | 0.2756              |
| Observations                                 | 10746               | 10746               | 10746               | 10746               | 18523               | 18523               | 18523               | 18523               |
| Individuals                                  | 1441                | 1441                | 1441                | 1441                | 2594                | 2594                | 2594                | 2594                |

Experience is modeled with a cubic polynomial. All equations control for year effects, education interacted with a cubic time trend, Black interacted with a cubic time trend, two-digit occupation at first job, and urban residence. Columns (1)–(4) control for sibling’s gender and the log of sibling’s wage interacted with a cubic time trend. Columns (5)–(8) control for father’s education interacted with a cubic time trend. For these time trends, the base year is 1992. For the models in columns (1) and (5), the coefficients on log of sibling wage and father’s education are .1600 and .0357, respectively, when evaluated for 1983. Standard errors are White/Huber standard errors computed accounting for the fact that there are multiple observations for each worker.

Table II: sibling wage and father’s education as hard-to-observe correlates.

## 6.2 Table II: sibling wage, father’s education, and schooling

Read:

- Columns (1)–(4) use sibling wage as the hard-to-observe productivity correlate.
- The sibling-wage coefficient rises with experience: the interaction coefficient is 0.1796 in column (2) and 0.1571 in column (4).
- The education-experience coefficient falls when sibling wage interactions are included.
- Columns (5)–(8) use father’s education. Its interaction is positive but less precise.

**Economic message:** The learning result is not unique to AFQT. A worker’s family-background proxy, especially sibling wages, also becomes more strongly priced as experience accumulates.

**Interpretation:** Sibling wages are a powerful proxy because siblings share family background, ability components, neighborhood influences, and possibly labor-market networks. If employers do not observe those

components at hire, they become visible through performance over time. Father’s education is more distant from productivity, so its evidence is weaker, which is informative rather than contradictory.

### 6.3 Table III: all hard-to-observe productivity correlates together

Read:

- The table includes AFQT, sibling wage, and father’s education simultaneously.
- In column (2),  $\text{AFQT} \times \text{experience}$  is 0.0843 and  $\text{sibling wage} \times \text{experience}$  is 0.1194.
- $\text{Education} \times \text{experience}$  falls from 0.0005 in column (1) to -0.0269 in column (2).
- In columns (3)–(4), adding the  $z \times \text{experience}$  interactions reduces the magnitude of  $\text{Black} \times \text{experience}$  from -0.1500 to -0.0861.

**Economic message:** The multivariate evidence confirms the mechanism: several hidden productivity proxies gain wage importance with experience, while schooling loses weight.

**Interpretation:** This is the strongest education-related table because the authors include multiple  $z$  variables together. The results suggest that schooling initially aggregates information from several correlated background and ability measures. The race result remains more ambiguous: adding hidden productivity correlates partly reduces the experience slope of the race gap, but does not make race behave like education in the model.

TABLE III  
THE EFFECTS OF STANDARDIZED AFQT, FATHER’S EDUCATION, SIBLING WAGE, AND  
SCHOOLING ON WAGES  
Dependent Variable: Log Wage; Experience Measure: Potential Experience.  
OLS estimates (standard errors)

| Model:                                     | (1)                 | (2)                 | (3)                 | (4)                 |
|--------------------------------------------|---------------------|---------------------|---------------------|---------------------|
| (a) Education                              | 0.0505<br>(0.0118)  | 0.0832<br>(0.0151)  | 0.0563<br>(0.0120)  | 0.0780<br>(0.0155)  |
| (b) Black                                  | -0.1333<br>(0.0255) | -0.1296<br>(0.0257) | 0.0454<br>(0.0609)  | -0.0284<br>(0.0704) |
| (c) Standardized AFQT                      | 0.0792<br>(0.0145)  | -0.0206<br>(0.0361) | 0.0789<br>(0.0144)  | 0.0065<br>(0.0413)  |
| (d) Log of sibling’s wage                  | 0.1602<br>(0.0208)  | 0.0560<br>(0.0352)  | 0.1617<br>(0.0207)  | 0.0604<br>(0.0351)  |
| (e) Father’s education/10                  | 0.0362<br>(0.0356)  | 0.0154<br>(0.0963)  | 0.0385<br>(0.0354)  | 0.0295<br>(0.0968)  |
| (f) Education *<br>experience/10           | 0.0005<br>(0.0093)  | -0.0269<br>(0.0123) | -0.0035<br>(0.0094) | -0.0220<br>(0.0128) |
| (g) Standardized AFQT<br>* experience/10   |                     | 0.0843<br>(0.0285)  |                     | 0.0614<br>(0.0333)  |
| (h) Log of sibling wage *<br>experience/10 |                     | 0.1194<br>(0.0393)  |                     | 0.1151<br>(0.0393)  |
| (i) Father’s education *<br>experience/100 |                     | 0.0176<br>(0.0789)  |                     | 0.0055<br>(0.0794)  |
| (j) Black * experience/10                  |                     |                     | -0.1500<br>(0.0474) | -0.0861<br>(0.0570) |
| $R^2$                                      | 0.2991              | 0.3014              | 0.3002              | 0.3016              |

Experience is modeled with a cubic polynomial. All equations control for year effects, education interacted with a cubic time trend, Black interacted with a cubic time trend, AFQT interacted with a cubic time trend, father’s education interacted with a cubic time trend, sibling wage interacted with a cubic time trend, two-digit occupation at first job, and urban residence. Also included are sibling’s gender and dummy variables to control for whether father’s education is missing and whether sibling’s wage is missing, and interactions between these dummy variables and experience when experience interactions are included. Standard errors

Original Table III: AFQT, father’s education, sibling wage, and schooling.

### 6.4 Tables IV and V: training controls and wage growth

Read: Table IV:

- Adding current and cumulative training does not eliminate the negative education-experience coefficient.
- Current training has a negative coefficient around -0.11, while cumulative training has a positive coefficient around 0.18.

- AFQT and sibling wage interactions remain positive in columns (2) and (4).

**Read: Table V:**

- The dependent variable is wage growth.
- $\text{AFQT} \times \Delta \text{experience}/10$  is positive, around 0.30 in columns (2) and (4).
- Sibling wage  $\times \Delta \text{experience}/10$  is also positive, around 0.21.
- Education  $\times \Delta \text{experience}/10$  becomes more negative when the  $z$  interactions are included.

**Economic message:** Training is important, but it does not absorb the learning pattern. Wage growth regressions continue to show that hidden productivity correlates gain importance as experience accumulates.

**Interpretation:** The training results highlight why learning and human capital are difficult to separate. Employers may allocate training to workers they learn are productive, so training itself can be an outcome of learning. The negative current-training coefficient and positive cumulative-training coefficient suggest that the training variables are not simple measures of human capital alone. They are also tied to selection, timing, and employer beliefs.

TABLE IV  
THE EFFECTS OF STANDARDIZED AFQT, FATHER'S EDUCATION, SIBLING WAGE,  
SCHOOLING, AND TRAINING ON WAGES  
Dependent Variable: Log Wage; Experience Measure: Potential Experience.  
Training Measure: Predicted before 88, Actual After.  
OLS estimates (standard errors)

| Model:                                              | (1)                          | (2)                          | (3)                          | (4)                          |
|-----------------------------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
| (a) Education                                       | 0.0606<br>(0.0119)           | 0.0802<br>(0.0151)           | 0.0651<br>(0.0121)           | 0.0746<br>(0.0155)           |
| (b) Black                                           | -0.1159<br>(0.0265)          | -0.1135<br>(0.0267)          | 0.0241<br>(0.0616)           | -0.0028<br>(0.0722)          |
| (c) Standardized AFQT                               | 0.0334<br>(0.0150)           | -0.0199<br>(0.0363)          | 0.0338<br>(0.0150)           | 0.0102<br>(0.0420)           |
| (d) Log of sibling's wage                           | 0.1594<br>(0.0213)           | 0.0716<br>(0.0357)           | 0.1611<br>(0.0213)           | 0.0759<br>(0.0356)           |
| (e) Father's education/10                           | 0.0460<br>(0.0356)           | 0.0211<br>(0.0974)           | 0.0482<br>(0.0354)           | 0.0353<br>(0.0977)           |
| (f) Education *<br>experience/10                    | -0.0231<br>(0.0095)          | -0.0392<br>(0.0123)          | -0.0260<br>(0.0096)          | -0.0339<br>(0.0128)          |
| (g) Standardized AFQT *<br>experience/10            |                              | 0.0460<br>(0.0287)           |                              | 0.0207<br>(0.0339)           |
| (h) Log of sibling's wage *<br>experience/10        |                              | 0.1041<br>(0.0402)           |                              | 0.1001<br>(0.0402)           |
| (i) Father's education *<br>experience/100          |                              | 0.0205<br>(0.0803)           |                              | 0.0084<br>(0.0805)           |
| (j) Black * experience/10                           |                              |                              | -0.1180<br>(0.0476)          | -0.0945<br>(0.0583)          |
| (k) Training: $R_t$                                 | -0.1143<br>(0.0200)          | -0.1095<br>(0.0199)          | -0.1115<br>(0.0199)          | -0.1091<br>(0.0199)          |
| (l) Cumulative training: $\Sigma$<br>$R_t$<br>$R^2$ | 0.1881<br>(0.0139)<br>0.3188 | 0.1830<br>(0.0139)<br>0.3199 | 0.1854<br>(0.0139)<br>0.3195 | 0.1827<br>(0.0139)<br>0.3202 |

Experience is modeled with a cubic polynomial. All equations control for year effects, education interacted with a cubic time trend, Black interacted with a cubic time trend, AFQT interacted with a cubic time trend, father's education interacted with a cubic time trend, sibling wage interacted with a cubic time trend, two-digit occupation at first job, and urban residence. Also included are dummy variables to control for whether father's education is missing and whether sibling's wage is missing, and interactions between these

Original Table IV: wage levels with training controls.

## 6.5 Tables VI and VII: sample composition and descriptive statistics

**Read: Table VI:**

- The final sample contains 2,976 individuals from NLSY79.
- Birth cohorts from 1957 to 1964 dominate the sample.
- Average education is 12.90 years and average number of observations is 7.08.

TABLE V  
ESTIMATES OF THE EFFECTS OF AFQT, FATHER'S EDUCATION, SIBLING WAGE,  
AND SCHOOLING ON WAGE GROWTH WITH CONTROLS FOR TRAINING  
Dependent Variable:  $\Delta \log$  Wage; Experience Measure: Potential Experience.  
Coefficient estimates (standard errors)

| Model:                                | (1)                 | (2)                 | (3)                 | (4)                 |
|---------------------------------------|---------------------|---------------------|---------------------|---------------------|
| Education *                           | -0.0060<br>(0.0833) | -0.0694<br>(0.0960) | -0.0106<br>(0.0832) | -0.0729<br>(0.0959) |
| $\Delta \text{experience}/10$         |                     | 0.3025<br>(0.1613)  |                     | 0.2975<br>(0.1614)  |
| AFQT * $\Delta \text{experience}/10$  |                     |                     |                     | 0.2107<br>(0.1477)  |
| Log of sibling wage *                 |                     | 0.2153<br>(0.1477)  |                     | 0.2107<br>(0.1477)  |
| Father's education *                  |                     | -0.4306<br>(0.5034) |                     | -0.4215<br>(0.5034) |
| $\Delta \text{experience}/10$         |                     | -0.0425<br>(0.0485) | -0.0503<br>(0.0483) | -0.0426<br>(0.0484) |
| Black * $\Delta \text{experience}/10$ | -0.0504<br>(0.0484) |                     |                     | 0.2468<br>(0.2429)  |
| Training: $R_t/10$                    |                     |                     | 0.1024<br>(0.1024)  | 0.1025<br>(0.1025)  |
| Lag training: $R_{t-1}/10$            |                     |                     | -0.0194<br>(0.1108) | -0.0230<br>(0.1108) |
| S.E.E.                                | .2965               | .2965               | .2965               | .2964               |

All equations control for year effects education, AFQT, sibling wage, and father's education all interacted with the change in the square and cube of time, the change in urban residence, and dummy variables to control for whether father's education is missing and whether AFQT is missing, and interactions between these dummy variables and the change in experience when change in experience interactions are included. Standard errors are White/Huber standard errors computed accounting for the fact that there are multiple observations for each worker. The sample size is 14,938 observations from 2703 individuals.

Original Table V: wage growth with training controls.



**Read:** Table VII:

- Mean real hourly wage is 8.23; mean log wage is 1.99.
- Mean potential experience is 7.09; mean actual experience is 4.74.
- Average schooling is 12.75 years, and 29 percent of observations are for Black workers.
- The standardized AFQT mean is -0.14 with a standard deviation of 1.04.

**Economic message:** These tables clarify the identifying sample: young workers observed repeatedly during early careers, with enough variation in schooling, race, AFQT, and family background to study learning.

**Interpretation:** The sample is designed for early-career employer learning, not for lifetime earnings analysis. The sibling-wage and father's-education variables have substantial missingness, which is why the authors include missing-data indicators and why the sibling specifications have smaller samples.

TABLE VI  
NLSY79 NON-HISPANIC MALE SAMPLE USED IN ANALYSIS, BY BIRTH YEAR

| Birth year | Number in<br>NLSY79 | Number in<br>sample | Percent in<br>sample | Years of<br>education | Ave. number<br>of obs. |
|------------|---------------------|---------------------|----------------------|-----------------------|------------------------|
| 57         | 743                 | 230                 | 31.0%                | 14.26                 | 7.19                   |
| 58         | 740                 | 272                 | 36.8%                | 13.93                 | 7.12                   |
| 59         | 715                 | 328                 | 45.9%                | 12.88                 | 7.42                   |
| 60         | 711                 | 433                 | 60.9%                | 12.77                 | 7.88                   |
| 61         | 615                 | 416                 | 67.6%                | 12.80                 | 7.58                   |
| 62         | 690                 | 494                 | 71.6%                | 12.63                 | 6.97                   |
| 63         | 646                 | 436                 | 67.5%                | 12.36                 | 6.59                   |
| 64         | 541                 | 366                 | 67.7%                | 12.57                 | 5.86                   |
| 65         | 2                   | 1                   | 50.0%                | 15.00                 | 9.00                   |
| Total      | 5403                | 2976                | 55.1%                | 12.90                 | 7.08                   |

Original Table VI: sample by birth year.

TABLE VII  
DESCRIPTIVE STATISTICS

| Variable                              | Mean  | Standard<br>deviation | Minimum | Maximum |
|---------------------------------------|-------|-----------------------|---------|---------|
| Real hourly wage                      | 8.23  | 4.71                  | 2.01    | 96.46   |
| Log of real hourly wage ( $w$ )       | 1.99  | 0.47                  | 0.70    | 4.57    |
| Potential experience ( $t$ )          | 7.09  | 3.61                  | 0       | 20.00   |
| Actual experience ( $t$ )             | 4.74  | 3.39                  | 0       | 14.92   |
| Education ( $s$ )                     | 12.75 | 2.13                  | 8.00    | 18.00   |
| Black dummy (Black)                   | 0.29  | 0.45                  | 0       | 1       |
| Standardized AFQT Score (AFQT)        | -0.14 | 1.04                  | -2.78   | 1.92    |
| Do not know sibling's wage            | 0.49  | 0.50                  | 0       | 1       |
| Log of sibling's wage                 | 1.95  | 0.45                  | 0.73    | 3.47    |
| Do not know father's education        | 0.12  | 0.33                  | 0       | 1       |
| Father's education                    | 11.68 | 3.28                  | 4.00    | 20.00   |
| Training ( $R_t$ )                    | 0.10  | 0.20                  | 0       | 1       |
| Cumulative Training: ( $\Sigma R_t$ ) | 0.48  | 0.40                  | 0       | 2.65    |

Original Table VII: descriptive statistics.

## 7 Short summary

- The paper studies employer learning and statistical discrimination in early-career wage setting.
- It predicts that initially observable traits used for screening should lose wage importance as employers learn.
- It predicts that initially hidden productivity correlates should gain wage importance with experience.
- NLSY79 data support this prediction for schooling versus AFQT, sibling wages, and father's education.
- The evidence for statistical discrimination based on race is weak and more difficult to interpret.
- Training controls do not overturn the main learning pattern.
- The paper's key contribution is a dynamic test of statistical discrimination based on changing wage coefficients, not static wage gaps.

## 8 Conclusion

### 8.1 Core conclusion

Altonji and Pierret show that wage coefficients can reveal information dynamics. If employers initially rely on easily observed credentials such as schooling, but then learn about worker productivity through job

performance, the wage return to schooling should fall with experience once hidden productivity correlates are allowed to enter with experience interactions. The evidence is consistent with this pattern.

## **8.2 Incremental scientific value**

The paper’s incremental value is methodological and substantive. Methodologically, it turns statistical discrimination from a static correlation problem into a dynamic coefficient problem. Substantively, it shows that schooling appears to serve partly as an initial information proxy, while AFQT and family-background measures become more strongly related to wages as workers age in the labor market.

## **8.3 Implications for education and signaling**

The results do not imply that education is merely a signal. They imply that part of the schooling wage premium reflects information that employers initially use but later replace with performance-based information. This interpretation weakens pure signaling stories if employers learn quickly, but it also weakens simple human-capital interpretations that treat schooling returns as time-invariant productivity effects.

## **8.4 Implications for discrimination**

The race results are important because they show how the same method can be used to ask whether employers use protected characteristics as information. The evidence is not consistent with full statistical discrimination on the basis of race in wages. However, the authors emphasize that race gaps may be driven by premarket skill formation, training access, networks, promotion, taste-based discrimination, or employment selection rather than wage-setting statistical discrimination alone.

## **8.5 Implications for firms and policy**

For firms, the paper suggests that better early productivity measurement could reduce inefficient reliance on broad observable proxies. For policy, it implies that banning certain variables from explicit use does not eliminate discrimination if employers can use correlated proxies. But it also shows that legal or ethical constraints may matter: race does not behave like education in the wage dynamics.

## **8.6 Research agenda**

The framework invites extensions to noncognitive skills, school quality, gender, immigrant status, neighborhood background, credit scoring, mortgage lending, and personnel records. The most valuable future data would combine initial employer information, direct performance evaluations, training assignments, promotions, and wage changes. Such data would make it possible to separate learning, human capital accumulation, and discrimination more cleanly.

## **8.7 Takeaway**

The paper’s enduring lesson is that wage dynamics contain information about information. When hidden productivity variables become more important and observable credentials become less important over time, the labor market is revealing a process of employer learning and statistical inference.